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Qu8its for Quantum Simulations of Lattice Quantum Chromodynamics

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We explore the utility of $d = 8$ qudits, qu8its, for quantum simulations of the dynamics of 1+1D SU(3) lattice quantum chromodynamics, including a mapping for arbitrary number of flavors and lattice size and a re-organization of the Hamiltonian for efficient time-evolution. Recent advances in parallel gate applications, along with the shorter application times of single-qudit operations compared with two-qudit operations, lead to significant projected advantages in quantum simulation fidelities and circuit depths using qu8its rather than qubits. The number of two-qudit entangling gates required for time evolution using qu8its is found to be more than a factor of five fewer than for qubits. We anticipate that the developments presented in this work will enable improved quantum simulations to be performed using emerging quantum hardware.

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I. INTRODUCTION

Quantum simulations of Standard Model physics are expected to provide results and insights about fundamental aspects of the structure and dynamics of matter that are not possible with experiment or with classical computing alone [1–9]. A major objective for high-energy physics and nuclear physics quantum simulations is to perform real-time simulations of non-equilibrium dynamics, such as of the low-viscosity quark-gluon liquid produced in heavy-ion collisions [10, 11], of the high-multiplicity events produced in proton-proton collisions [12], and the creation of matter in the early Universe [13]. Progress toward such simulations is currently at early stages, in terms of the capabilities of quantum computers, of the sophistication of relevant quantum algorithms and workflows, and in understanding how to co-design quantum computers to best simulate quantum field theories. A degree of focus is being placed on 1+1D systems, such as the Schwinger model, SU(3) quantum chromodynamics (QCD) and its SU(2) analog, and the Gross-Neveu model, in order to prepare for simulating QCD in 2+1D and 3+1D, with the ultimate goal of providing robust results (with a complete quantification of uncertainties) for observables that cannot be assessed in the laboratory or with observation.

Most of the development so far has been centered around quantum computers utilizing qubits [4, 11–53], such as trapped-ion systems and superconducting qubit systems. However, in pursuit of hybrid qubit-qudit architectures [54, 55], further device capabilities, such as vibrational excitations [56–60] or superconducting radio-frequency (SRF) cavities [22, 61–63], are being integrated into system functionalities. Compared to qubits alone, quantum simulations using higher-dimensional qudits [64],[1] which can allow for "better fit" Hilbert spaces and reductions in the number of entangling gates [66], take us, in a sense, one more step away from classical computing. With recent developments in quantum hardware, including results from qudit trapped-ion systems [67–70], superconducting circuits [71–73], photonic systems [74], and nitrogen vacancy centers in diamond [75], and anticipating that such devices will become increasingly capable and available (in particular, trapped-ion systems with $d > 10$ [69]), a more general consideration of how one can utilize qudits in simulations of Standard Model quantum field theories is timely. Examples of such applications can be found in Refs. [22, 54, 55, 76–83], where the larger Hilbert space of qudits is used to describe the gauge fields of (non-)Abelian lattice gauge theories, or in Refs. [66, 84], where nuclear many-body systems naturally map to qudits.

Quantum simulations of 1+1D SU(2) [11, 18, 23, 24, 34, 51] and SU(3) [22, 30, 31, 37–39, 44, 53] lattice gauge theories have recently been performed. Those that included matter fields have used the Kogut-Susskind (KS) staggered discretization[2] of the quark fields [89, 90], and worked in axial gauge to utilize Gauss's law to uniquely define the gauge fields [37, 91], enabling their contributions to be included by non-local all-to-all interactions. In this mapping, one color of one flavor of quark is mapped to two qubits, one describing the occupation of that quark and one of the corresponding anti-quark. For example, to describe one site of two-flavor ($N_f = 2$) SU(3) QCD requires 12 qubits.[3] Quantum circuits for preparing the ground state and implementing Trotterized time evolution have been identified and the quantum resources established (providing an upper bound) [37].

In this work, we explore the utility of qudits with $d = 8$, which we denote as qu8its,[4] in simulating 1+1D lattice QCD using the KS discretization. This is motivated, in part, by our demonstration of the utility of using qu5its in quantum simulations of multi-fermion (nucleon) systems with pairing interactions [66] and an underlying SO(5) symmetry. The 8 states associated with a single quark flavor mapped to three qubits can be mapped to the states of a single qu8it, and the 8 states associated with single anti-quark mapped to three qubits can be mapped to another qu8it (which we loosely denote as an anti-qu8it). While it does not seem to be an immediate gain by counting the number of states, the advantage is in the number of entangling gates required to implement time evolution, and in the number of units in a quantum register. Analogous to qubit operations, the single-qudit gate operations are significantly faster and of higher fidelity than the two-qudit operations.[5] Consequently, the factor of $\gtrsim 5$ reduction in the number of two-qudit gates and the factor of 3 reduction in the number of units in the register that we find in mapping to qu8its, suggest that quantum computers with qu8its are likely to provide enhanced capabilities for simulating non-Abelian lattice gauge theories.

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[1] For a recent review of qudits, see, for example, Ref. [65].
[2] Different discretization approaches, such as Wilson fermions, are also being investigated [25, 85–88].
[3] The lepton fields have also been included, in simulating the β-decay of a baryon [39], requiring four additional qubits per lepton generation.
[4] Which we suggest is pronounced *q-huits*.
[5] For example, in IonQ's current flagship trapped-ion qubit quantum computer, single-qubit operations require $\sim 100\ \mu s$ while two-qubit operations require $\sim 600\ \mu s$ [92].